

HumidClimatologyEngine

v6 vs v7.5 - Scientific and Engineering Contract

Hourly empirical moisture climatology, five-day centred fitting windows, multi-family model selection, empirical 2-D PDF, copula candidates, and power-failure-safe checkpointing

Release note: v7.5 is the new hourly empirical production line; v6 remains as a historical/educational baseline.

1. Executive decision

v6 is retained but is not the preferred production method. Its daily-statistical joint-Gaussian/Monte-Carlo pipeline is useful for teaching covariance, uncertainty propagation and PSD validation. v7.5 starts from hourly ERA5-Land states and preserves the time-scale information needed for direct empirical moisture distributions.

Dimension	v6	v7.5
Input	Daily statistics / daily fields	Hourly observations
Primary moisture statistics	Gaussian-MC transformed	Direct empirical RH/e/r/q moments
Distribution candidates	Joint Gaussian state	Normal, Skew-Normal, Pearson III, Beta, Bimodal Normal
2-D probability	Gaussian candidate	Empirical 2-D PDF + copula candidates
Fitting window	Not central to production fit	Centered 5-day window, D-2..D+2
Restart unit	Annual/day state	DOY x spatial chunk; transactional commit
Padding	Not part of hourly window contract	1980-12-30/31 + 2021 data used as temporal padding only

2. v6: what was wrong and why it is still worth keeping

Hourly ERA5-Land
-> daily reduction
-> daily (T, Td, lnP) moments/covariance
-> joint Gaussian model
-> Monte Carlo
-> RH / e / r / q

The key scientific limitation is the early loss of hourly timing and within-day co-variation. A later covariance model cannot reconstruct information that has already been collapsed by daily aggregation. The use of Gaussian/Monte-Carlo is therefore a second methodological layer, not the sole reason v6 is not the production baseline.

v6 should remain documented as a **historical and educational method**. Its covariance, PSD repair and stochastic-propagation components are useful reference implementations.

3. v7.5 primary architecture

Hourly T2m + D2m + SP
|
actual timestamp alignment
|
thermodynamic transformation
|
empirical online moments
|
DOY climatology + diagnostics
|
empirical 2-D PDF
|
separate five-day fitting/query layer

The primary climatology remains empirical and direct. The five-day layer is deliberately separated so expensive distribution fitting and station queries do not destabilize the restartable accumulation engine.

4. Five-day centred window contract

For every target date D , the distribution fitting sample is taken from the complete five-day interval $D-2$ through $D+2$. The centre day is the label; neighbouring days enlarge the sample only for fitting.

D-2 | D-1 | D | D+1 | D+2
 $\wedge$
 target day

The window is constructed at the **raw timestamp layer**. It is never recreated after DOY-level aggregation. The loader reads the actual datetime coordinate inside NetCDF and therefore supports the demonstrated ``valid_time`` coordinate in the 1980 edge file.

For the 1981-2020 target climatology, 1980-12-30 and 1980-12-31 are padding for the first target windows. Data in 2021 are padding for the final target windows. Padding observations are not treated as additional climatology years.

The dedicated edge file is explicitly supported:

K:\kazemi\papers\temperature_interpolation\19801230-19801231.nc

Its ``valid_time`` coordinate is the source of truth. Surface-pressure coverage is tracked independently because the supplied edge file contains T2m and D2m but not SP; this matters for r and q and for any RH- q copula fit.

5. Distribution candidate family

v7.5 does not force a single family on all days and cells. Candidate fitting is performed for the extracted five-day sample:

Family	Use
Normal	Symmetric baseline
Skew-Normal	Continuous asymmetric states
Pearson Type III	Skewed continuous climate states
Beta	Bounded variables such as RH/100 and q
Bimodal Normal	Potentially two-regime / two-peak behaviour

6. Skew-Normal fitting from ClimateProcessingEngine

The v7.5 candidate set retains the public project logic as an auditable candidate: sample mean, standard deviation and skewness are used to construct ``(xi, omega, alpha)``, and the log-likelihood is then evaluated. A second Skew-Normal candidate uses scipy maximum likelihood so the same family can also be compared after likelihood refinement.

This avoids silently replacing the fitting convention that worked well in the temperature project.

7. Bimodal Normal: five parameters, not “double normal PDF”

The model is a two-component Gaussian mixture. Its five free parameters can be written as:

```
w1, mu1, sigma1, mu2, sigma2  
w2 = 1 - w1
```

v7.5 uses deterministic multi-start EM, with the project-aligned defaults ``n_init=10``, ``max_iter=1000``, ``tol=1e-4`` and ``reg_covar=1e-6``. Components are ordered by their means. The fit record keeps separation and overlap diagnostics in addition to AIC/AICc/BIC.

This is essential for moisture because RH, q and related variables can inherit regime mixtures from synoptic or air-mass regimes. A visually skewed distribution should not automatically be forced into a one-component skew family.

8. Model selection

```
fit all admissible candidates  
  |  
log-likelihood  
  |  
AIC / AICc / BIC  
  |  
multimodality diagnostics  
  |  
tail / boundary diagnostics  
  |  
stability checks  
  |  
selected model + full candidate table
```

AICc is used as the primary ranking statistic when enough observations exist; BIC is retained as a second criterion. The full candidate table remains stored, so “best model” is not a black box.

9. Two-dimensional probability

The empirical 2-D PDF is the primary scientific reference. It is built from observed hourly paired values and represented as a piecewise-constant density. A bivariate Gaussian surface is never assumed as the truth.

```
paired hourly samples  
  |  
rank / bin / empirical representation  
  |  
empirical joint PDF  
  |  
optional parametric marginal fits  
  |  
optional copula dependence model
```

For bounded variables, Beta is a candidate marginal. A copula is a dependence model, not a substitute for the marginal distribution. Gaussian, t and alternative copulas should be compared by fit and tail diagnostics before being accepted.

10. Professional chunking and station queries

The extraction layer follows the same separation of concerns demonstrated by ClimateProcessingEngine: monthly reading, spatial block selection, and cacheable data access are kept separate. The public project explicitly implements block-based monthly reading with disk/memory caching.

```
station / grid-cell request
|
identify intersecting months
|
read only spatial block
|
reuse disk/in-memory cache
|
align actual timestamps
|
5-day window
|
fit distributions / copula
```

This prevents a single station request from loading the complete 301 x 301 grid. It also makes repeated distribution fitting substantially more practical.

11. Power-failure-safe processing

The main 1981-2020 accumulation uses a transactional checkpoint unit of **DOY x spatial chunk**. A chunk is marked complete only after its sufficient statistics are written and the NetCDF checkpoint is synchronized. A power loss therefore reprocesses only uncommitted units.

```
compute chunk
-> write state
-> sync()
-> completed_chunk = 1
-> sync()
-> committed

power failure
-> resume unfinished units only
```

Progress is explicitly reported as percentage, completed/total units, remaining units, processing rate and ETA. This information is also persisted in JSON run metadata.

12. Validation status for v7.5

- Python syntax compilation passes.
- A synthetic two-mode sample selects Bimodal Normal as the best model under the implemented criteria.
- Bounded RH/q synthetic samples fit Beta candidates successfully.
- Gaussian copula fitting on pseudo-observations returns finite dependence parameters.
- The time-coordinate resolver accepts the demonstrated `valid\_time` coordinate.
- The full 1981-2020 ERA5-Land production run has not been claimed here because the user data filesystem is not mounted in this execution environment.

13. Final scientific position

v7.5 should be interpreted as an empirical hourly framework with a flexible distribution-selection layer. The correct workflow is not “choose a double normal.” It is “observe the data, fit admissible candidates, keep the empirical joint distribution as a reference, and let diagnostics determine whether a single-skew, bounded, or bimodal family is justified for each day and location.”